

# Chapter 7

## Pattern theorems

### 7.1 Patterns

In this chapter we shall prove a useful theorem due to Kesten (1963) about the occurrence of patterns on self-avoiding walks, and investigate a number of its applications. Briefly, a *pattern* is a (short) self-avoiding walk that occurs as part of a longer self-avoiding walk. Kesten's Pattern Theorem says that if a given pattern can possibly occur several times on a self-avoiding walk, then it must occur at least  $aN$  times on almost all  $N$ -step self-avoiding walks, for some  $a > 0$  (in this context, "almost all" means "except for an exponentially small fraction"). This can be viewed as a weak analogue of classical "large deviations" estimates for the strong law of large numbers, which say that certain events have exponentially small probabilities [see for example Chapter 1 of Ellis (1985)].

Another statistic of interest regarding patterns is the frequency of occurrence of a particular pattern at the beginning of self-avoiding walks. In general dimension  $d$ , it is an open problem to prove that the fraction of  $N$ -step self-avoiding walks that begin with a given pattern converges as  $N$  tends to infinity. This has been done in certain special cases: for  $d \geq 5$  (see Section 6.7), and for bridges in every dimension (see Section 8.3). The existence of such a limit would provide a natural definition of a probability measure for infinite self-avoiding walks. We can only prove the following weaker results in the general case: if  $P$  is a pattern that can occur at the beginning of an arbitrarily long self-avoiding walk, then the fraction of  $N$ -step self-avoiding walks beginning with this pattern is bounded away from zero as  $N$  tends to infinity; also, the ratio of these fractions for  $N$  and  $N+2$  converges to one. These results and some extensions will be discussed in

Section 7.4. The proofs of these results rely heavily upon Kesten's original pattern theorem.

Kesten originally applied his pattern theorem to prove the following *ratio limit theorems*:

$$\lim_{N \rightarrow \infty} \frac{c_{N+2}}{c_N} = \mu^2, \quad (7.1.1)$$

$$\lim_{N \rightarrow \infty} \frac{q_{2N+2}}{q_{2N}} = \mu^2, \quad (7.1.2)$$

$$\lim_{N \rightarrow \infty} \frac{b_{N+1}}{b_N} = \mu. \quad (7.1.3)$$

We shall prove these results in Section 7.3. Unfortunately, the same methods do not allow us to prove

$$\lim_{N \rightarrow \infty} \frac{c_{N+1}}{c_N} = \mu. \quad (7.1.4)$$

Equation (7.1.4) in  $\mathbf{Z}^d$  has only been proven for  $d \geq 5$  (see Theorem 6.1.1); finding a proof for  $d = 2, 3, 4$  remains an open problem. To get a feeling for why (7.1.1) is easier to prove than (7.1.4), consider the following easy exercise: prove that

$$c_{N+2} \geq c_N \quad \text{for every } N. \quad (7.1.5)$$

The idea of the proof is given in Figure 7.1. (In detail: Given an  $N$ -step

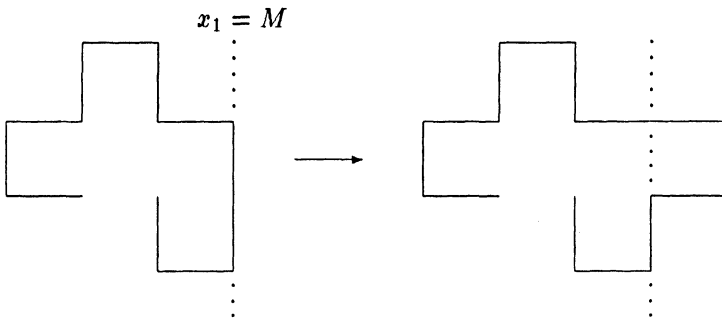


Figure 7.1: The idea behind the proof that  $c_{N+2} \geq c_N$ : increasing the length of a self-avoiding walk by 2.

self-avoiding walk  $\omega$ , let  $M = \max\{\omega_1(i) : 0 \leq i \leq N\}$ . On the one hand, if a step of  $\omega$  joins two points  $u$  and  $v$  in the hyperplane  $x_1 = M$ , then replace